SUMIT SINGH

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OBJECTIVE

AI/ML engineer experienced in language-model fine-tuning, retrieval, and production conversational systems. Seeking research and engineering opportunities to investigate model architectures, representation learning, and training methods.

EDUCATION

Master of Science in Applied Computing, *AI Concentration* Sep 2026 - Dec 2027
University of Toronto
Relevant Coursework: Neural Networks and Deep Learning; Imitation Learning for Robotics

Bachelor of Technology, Computer Science Engineering *(Specialization in AI & ML)* Jul 2018 - May 2022
SRM Institute of Science and Technology CGPA - 9.71/10.00

PROFESSIONAL EXPERIENCE

Goldman Sachs — Associate (SDE-2) Jul 2022 - Jul 2026

• **Conversational AI-based IVR:** Developed and deployed an IVR using **Amazon Lex**, **AWS Lambda (Go)**, and Java microservices, replacing DTMF menus with conversational support across **60+ customer intents**; upgraded Lex from ASR 1.0 to **ASR 2.0** to improve transcription and intent classification for customer requests expressed in everyday language.
• **Intent Classification & Fine-tuning:** Fine-tuned **DistilBERT and Llama** on customer utterances, testing **LoRA/QLoRA** for Llama and model quantization; selected DistilBERT for its balance of accuracy and inference speed on internal evaluations.
• **Customer Self-Service:** Improved the IVR's ability to resolve customer requests without agent transfers, contributing to containment growth from **10% to 75%** across **600K monthly calls** and **$5M+ in quarterly savings**.
• **RAG for Agent Assistance:** Developed a **speech-to-text RAG** workflow to generate answer suggestions from FAQs and documentation, covering knowledge-base preparation and inference evaluation of retrieval relevance and answer quality.
• **Call Summarization:** Helped integrate **Amazon Bedrock** into the call workflow and refined prompts for transcript summaries that gave agents context at handoff and helped teams analyze recurring customer pain points to improve bot workflows.
• **Automated Bot Regression Testing:** Engineered a serverless workflow using **AWS Lambda and Step Functions** to process **3,000+ test utterances weekly**, generating reports on model and bot issues to identify regressions during bot updates.

Sabre — Associate Intern Jan 2022 - Jul 2022

• **Automated Python Data Scraping:** Developed Python-based web scraping scripts to collect airline data from public websites, and prepared the data in a structured format for downstream analysis with minimal manual intervention.
• **Airline Data Analytics:** Prepared and formatted airline data for an internal analytics and optimization tool, supporting analysis of flight scheduling, fleet allocation, and dynamic pricing for client airlines, with team and client reviews of the outputs.

PERSONAL PROJECTS

• **RF Capture Auto-labeler — Qoherent (Ongoing):** Developing an auto-labeling workflow for radio-frequency captures. Reproduced supplied signal-detection and classification baselines on synthetic data, inspecting training and validation behavior to guide experiments on detection quality and unknown-signal rejection.

• **PhysicsTeacher-CurrentElectricity-GPT** 🔗: Built a physics tutoring system for Grade 11 Current Electricity using **RAG and ChromaDB**, with textbook retrieval and **embedding-based conversation memory**. Used **OpenAI** generation and **NVIDIA NeMo Guardrails** for grounded, domain-constrained responses and context-aware follow-up handling.

• **ML-based Invoice Management Application** 🔗: Preprocessed **45K+** invoices, engineered **6 payment-trend features**, compared models, and tuned XGBoost via **5-fold cross-validation**, yielding **2.43-day MAE**, **0.61 R^2**.

KEY SKILLS

Languages: Python, Go, C++, Java, SQL, JavaScript
ML/AI: Fine-tuning (LoRA/QLoRA), Quantization, NLP, RAG, Prompt Engineering, scikit-learn, XGBoost
Tools: PyTorch, Hugging Face Transformers, ChromaDB, NVIDIA NeMo Guardrails, Amazon Lex, Bedrock, AWS Lambda

EXTRA-CURRICULAR ACTIVITIES

Authored **8 project tutorial blogs** on **CodeSpeedy.com**, generating **11,000+** cumulative views. 🔗